

Immersive Chemistry Lab

Testing Plan for Interactive Prototype 1

DECO2300/7230 Digital Prototyping and Extended Reality

3-Sentence Project Pitch

Immersive Chemistry Lab is an XR redesign of the ChemCollective Virtual Lab that allows students to interact with chemistry at both the visible laboratory scale and the molecular scale. Instead of only observing a reaction on a screen, users can manipulate reactants, trigger a reaction, transition into a molecular representation, and directly construct and manipulate the resulting molecule. The main idea is to use spatial XR interactions to connect what happens during a visible experiment with what happens at the molecular level.

Testing Objective

The main unknown I want to test is whether the interaction sequence is understandable to a new user and whether the transition between the laboratory and molecular views feels like one connected experience rather than two unrelated environments.

My initial concept assumed that users would understand that they can interact directly with objects in 3D space, recognise the scale-changing interaction, and understand that the molecular environment represents what is happening within the laboratory reaction. These assumptions have not yet been properly validated with the interactive Unity prototype.

This test therefore focuses on three main questions:

Testing question	What I want to learn
Can users understand how to progress through the reaction without excessive instruction?	Whether the reactant manipulation and reaction feedback make the first stage understandable.
Do users understand the transition from the visible reaction to molecular scale?	Whether the focus object and spatial controller movement communicate that the user is moving 'inside' the reaction rather than simply loading another scene.
Do users understand what they can do with the molecule?	Whether grabbing Hydrogen atoms, forming bonds, and pulling atoms apart are discoverable and whether the interactions communicate molecular construction and bond breaking.

I am not using this prototype to measure whether users learn chemistry better in VR. The prototype is too early for that claim. The focus of IP1 is whether the interaction model itself is understandable enough to justify developing the concept further.

Testing Methodologies

I will use a moderated, task-based usability test with think-aloud observation, followed by a short semi-structured interview.

Participants will be given goals rather than step-by-step instructions. For example, instead of telling them exactly how to move a Hydrogen atom, I will ask them to try to complete the molecule. This is important because giving away the interaction beforehand would prevent me from finding out whether the interaction itself is understandable.

During each task I will ask participants to briefly say what they think is happening and what they are trying to do. I will avoid correcting them immediately. If they become completely stuck, I will provide a small prompt and record that assistance was required.

I chose this approach because the most useful evidence at this stage is where users hesitate, what movements they naturally attempt, what they misunderstand, and whether the physical interaction matches their expectations.

Prototype Description / Requirements

Laboratory-scale section

The user begins at a small virtual laboratory bench containing two reactants and a reaction beaker. The reactants can be spatially moved towards the reaction area. Once both reactants are detected, the prototype provides reaction feedback through a particle effect and activates a molecular focus object above the reaction.

Scale-transition section

After the reaction occurs, a molecular focus appears above the beaker. The user performs a spatial expansion/controller movement interaction to initiate the transition. The laboratory focus then disappears and the molecular environment becomes visible.

The intention of this interaction is to represent the user moving from the observable reaction into its molecular representation.

Molecular-scale section

At molecular scale, the user can:

- grab Hydrogen atoms;
- move them spatially towards Oxygen;
- snap them into valid bonding positions;
- see a bond appear when the atom connects;
- grab a bonded Hydrogen again and pull it away;
- observe the bond stretch as distance increases;
- break the bond by pulling the atom far enough away.

The prototype is intentionally horizontally focused. It demonstrates the main experience and interaction sequence rather than implementing a complete chemistry simulation.

Current prototype limitation

The prototype is currently being tested through Unity's XR Interaction Simulator rather than as a fully finished headset implementation. Some interactions therefore use simulated controller input and are simplified representations of the final physical gestures.

This limitation should be considered when interpreting results. I am primarily testing whether users understand the interaction concept and sequence, not the final ergonomics or physical accuracy of VR hand/controller tracking.

Assumptions Being Tested

Assumption	Evidence that would support it	Evidence that would challenge it
Users understand that the reactants should be brought into the reaction area.	Participants attempt to move the reactants towards the beaker with little or no prompting.	The participant waits, searches for buttons, or does not associate the reactants with the beaker.
Reaction feedback clearly communicates that something has happened.	The participant notices the visual reaction and recognises that the experiment has progressed.	The participant does not notice the reaction or continues trying the same action.
Users understand that the molecular focus represents a transition into the reaction.	Participants associate the focus/transition with exploring the reaction more closely.	Participants described it as an unrelated object, menu, or teleport.
The spatial scale-change interaction is understandable.	Participants attempt an outward/spatial controller movement without being explicitly shown.	The participant searches for a button or cannot identify how to continue.
Users recognise that molecular objects are directly manipulable.	The participant reaches towards or attempts to grab Hydrogen.	The participant only observes the molecule or looks for UI controls.
Snapping communicates molecule construction.	Participants understand that bringing Hydrogen close to Oxygen forms a bond.	Participants believe the snap was accidental or do not understand the bond visual.
Stretching and breaking communicates bond manipulation.	The participant intentionally pulls the atom away and understands why the bond stretches/breaks.	The participant interprets the disappearing bond as a bug or does not connect distance with breaking.
The two scales feel connected.	Participants can explain that the molecular environment represents the earlier reaction.	Participants experienced them as two separate or unrelated activities.

Data Collection Method

For each participant I will use a simple observation sheet rather than relying on memory after several tests.

Measure	How it will be recorded
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Task completion	Completed independently / completed with prompt / not completed
First action attempted	Short note describing what the participant naturally tried first
Hesitation	Whether there was an obvious pause or uncertainty before acting
Errors / unexpected actions	What the participant attempted that did not progress the interaction
Help required	Number/type of prompts given by the facilitator
Interpretation	What the participant said they thought the interaction represented
Think-aloud comments	Short notes of particularly useful comments
Overall connection between scales	Whether they correctly connected the molecular environment with the laboratory reaction

I will also record approximate completion time, but speed will not be treated as the main measure of success. A participant taking slightly longer while correctly understanding the interaction is more useful than completing quickly after being told exactly what to do.

Success / Validation Criteria

I will consider the interaction model provisionally validated if most participants can progress through the main interaction sequence with no more than minor prompting and correctly understand the relationship between the two scales.

Criterion	Target
Participants complete the reaction stage	At least 4/5 with no more than one minor prompt
Participants initiate the molecular transition	At least 3/5 without being explicitly shown the gesture
Participants successfully form at least one molecular bond	At least 4/5 with no more than one minor prompt
Participants discover or understand the pull-to-break interaction	At least 3/5 without a full demonstration
Participants can explain that molecular mode is connected to the laboratory reaction	At least 4/5
Major recurring confusion	No single critical interaction should fail for the majority of participants

I am deliberately setting a slightly lower threshold for discovering the scale gesture and bond-breaking interaction because these are the more unusual interactions and therefore the areas where testing is most valuable.

A failure against these criteria would not mean the whole concept has failed. It would identify where stronger visual cues, interaction feedback, onboarding or gesture design are required for the next prototype.

Testing Setup

1. Open the correct Unity scene and reset the prototype so the reactants and molecular objects are back in their starting positions.
2. Check that the XR Interaction Simulator and both controllers are responding.
3. Check that Reactant A and B can trigger the reaction.
4. Confirm that MolecularWorld starts hidden and that MolecularFocus only appears after the reaction is triggered.
5. Confirm that Hydrogen grabbing, snapping, bond stretching and breaking work.
6. Prepare the observation sheet before starting the participant.
7. Explain that this is an early prototype and that I am testing the design, not the participant.
8. Ask the participant to think aloud and say what they expect to happen while interacting.

I will avoid explaining the specific gestures before the corresponding task unless assistance becomes necessary.

Testing Process - approximately 5 minutes per participant

1. Introduction - approximately 30 seconds

Participant introduction: "This is an early prototype of an immersive chemistry lab. Imagine you are using VR controllers to perform a chemistry experiment and explore what happens at molecular scale. I'm testing whether the interactions make sense, not whether you know chemistry. Please say what you are thinking while you use it."

Explain the basic simulator/controller controls needed to operate the prototype, but do not explain the chemistry-specific interaction sequence.

2. Task 1 - Start the reaction - approximately 45 seconds

Participant instruction: "You have two reactants and a reaction beaker in front of you. Try to start the experiment."

Observe which object they approach first, whether they attempt to grab/move the reactants, whether the beaker is understood as the destination, whether help is required, and whether they notice the reaction feedback.

After the reaction, ask: "What do you think just happened?"

3. Task 2 - Enter molecular scale - approximately 45-60 seconds

Once the molecular focus appears, say: "You now want to investigate what is happening inside this reaction. Show me what you would try next."

Do not initially explain the expansion gesture. Observe whether the participant approaches the focus, what movement they try first, whether they naturally use one or both controllers, hesitation before acting, whether they search for a button instead, and whether a prompt is required.

If stuck, use progressively stronger prompts:

Prompt 1: "Try interacting with the object spatially."

Prompt 2: "Try moving your controller(s) around or away from the focus."

Only demonstrate the intended action if they still cannot proceed, and record this as full assistance.

After transition, ask: “What do you think this new environment represents?”

4. Task 3 - Construct the molecule - approximately 60 seconds

Participant instruction: “Try interacting with the atoms and completing the molecule.”

Observe whether they attempt to grab Hydrogen, understand spatial positioning, move Hydrogen toward Oxygen, react to the snapping behaviour, understand the visible bond, and require prompting.

After a successful snap ask: “What do you think happened when the atom connected?”

5. Task 4 - Manipulate the bond - approximately 45 seconds

Participant instruction: “Now see what happens if you manipulate the molecule you just created.”

Avoid saying “pull the Hydrogen until the bond breaks” initially. Observe whether they grab the bonded Hydrogen again, rotate or pull it, notice the bond stretching, whether stretching encourages them to continue pulling, and their reaction when it breaks.

Afterwards ask: “Why do you think the bond disappeared?”

6. Short post-test questions - approximately 60 seconds

Ask:

- Which interaction felt most natural?
- Which interaction was the least obvious or most confusing?
- What did you think the transition from the lab to the molecule represented?
- Without me explaining it now, what do you think you were doing when you connected and separated the atoms?
- Was there anywhere you expected feedback or an instruction but didn't get one?
- If time permits: What is one thing you would change to make the experience clearer?

Observation Sheet

Use one row per participant. I = Independent, P = Prompt required, N = Not completed.

Participant	Reaction	Scale transition	Form bond	Break bond	Prompts needed	Main hesitation / error	Understood lab → molecular connection?	Key comment
P1	I / P / N	I / P / N	I / P / N	I / P / N			Yes / Partly / No	
P2	I / P / N	I / P / N	I / P / N	I / P / N			Yes / Partly / No	
P3	I / P / N	I / P / N	I / P / N	I / P / N			Yes / Partly / No	
P4	I / P / N	I / P / N	I / P / N	I / P / N			Yes / Partly / No	
P5	I / P / N	I / P / N	I / P / N	I / P / N			Yes / Partly / No	

Basis for This Testing Plan

This plan follows the IP1 brief structure and develops the initial testing direction from the Design Concept Report. The focus remains on testing understandability of the interaction model, especially the reaction sequence, transition between laboratory and molecular scale, and direct molecular manipulation. The plan is designed for short in-class sessions of approximately five minutes per participant and for collecting actionable observations that can inform the next prototype iteration.